How Do Annual Members and Casual Riders Use Cyclistic Bikes Differently?

Rafael Couto · October 2026 · Tools: Excel, Google BigQuery (SQL), Google Sheets

Executive summary

According to the case brief, Cyclistic's finance analysts have concluded that annual members are more profitable than casual riders. This analysis takes that as a premise, since the trip data contains no financial information to verify it. Using 5.9 million rides (September 2025 – August 2026), it shows how the two groups use the service differently, to support a marketing strategy that converts casual riders into members.

Key findings

- Members take 65% of rides and ride in a commuting pattern: weekdays, with peaks at 8 AM and 5 PM.
- Casual riders take 35% of rides, but their rides last ~48% longer (17.8 vs. 12.0 minutes). They peak on weekends and are far more seasonal: their rides fall 14x from July to January, versus 5x for members.
- Yet 63% of casual rides happen on weekdays, and between 144K and 526K of them fit a commuting pattern. On weekday mornings, many start next to Metra commuter-rail terminals and in residential and office areas.

Recommendations

1. Target casual riders who already commute, at Metra terminals and business districts, with digital in-app messages.
2. Run the conversion campaign from spring to early fall, ahead of the winter drop.
3. Position membership as one pass for both commuting and leisure, aimed at weekend casual riders.

1. Ask

Business task

Analyze how annual members and casual riders use Cyclistic bikes differently, using the last 12 months of historical trip data, to inform a marketing strategy aimed at converting casual riders into annual members, the company's most profitable customer segment.

Lily Moreno, Director of Marketing, believes converting existing casual riders is more promising than acquiring new customers, because they already know and use Cyclistic. This analysis provides the evidence to design that campaign: who to target, with what message, when and where.

Stakeholders

Stakeholder	Role and interest
Lily Moreno, Director of Marketing	Set the goal; will use the insights to design the conversion campaign.
Cyclistic executive team	Approves or rejects the recommendations. Detail-oriented: needs compelling, well-supported insights and clear visuals.
Marketing analytics team	Collects, analyzes and reports data that guides marketing strategy (the analyst's team).

2. Prepare

Data source

Cyclistic is a fictional company; the data is real public trip data from Divvy, Chicago's bike-share system, made available by Motivate International Inc. under the Divvy Data License Agreement. Period analyzed: September 2025 – August 2026 (12 monthly files).

Organization

Each monthly CSV file contains one row per ride, with 13 columns. The original files were stored untouched in a dedicated raw-data folder; all work was done on copies and in BigQuery.

Column(s)	Description
ride_id	Unique identifier of the ride
rideable_type	electric_bike or classic_bike
started_at, ended_at	Start and end date/time of the ride
start/end_station_name, _id	Start and end station (blank for e-bikes parked outside a station)
start/end_lat, _lng	Start and end coordinates
member_casual	Rider type: member (annual) or casual (single ride / day pass)

Credibility (ROCCC)

Criterion	Assessment	Justification
Reliable	Yes, with checks	System-generated by the operator; contains system errors (false starts, unreturned bikes) handled in cleaning.
Original	Yes	First-party data published by the operator.
Comprehensive	Partially	No rider ID, price, demographics or pass type (see limitations).
Current	Yes	Most recent 12 months (Sep 2025 – Aug 2026).
Cited	Yes	Public source with a published license.

Privacy and limitations

- No personally identifiable information is included: there is no rider or payment-card ID. This protects privacy, but it means the analysis compares rides, not people. It is not possible to know how many unique riders are behind the rides, or how often each one rides.
- No pricing data: membership savings for a given rider cannot be calculated from this dataset.
- No pass type: casual rides mix single rides and day passes.
- Rider behavior (commuting vs. leisure) is inferred from ride patterns (day, hour, duration, station), not observed directly.
- Profitability by rider type is taken from the case brief and could not be verified with this dataset.

3. Process

Tools and why

- **Excel:** explored one month (September 2025) to understand the data, test quality checks and form hypotheses.
- **Google BigQuery (SQL):** the full year has over 6 million rows, beyond Excel's limit of ~1 million per sheet. The 12 files were loaded into BigQuery, combined with UNION ALL and cleaned with SQL.
- **Google Sheets:** visualizations built from the SQL query results.

Data quality checks

Check (raw combined table)	Result	Action
Total rows (12 files combined)	6,115,982	Starting point
Rides starting outside the analysis period (Aug 2025)	268	Removed
Duplicate ride_id across monthly files	35	Kept one record
Rides shorter than 1 minute	158,904 (2.6%)	Removed
Rides of 12 hours or more	7,558 (0.12%)	Removed
Blank start station name	1,285,406 (21%)	Kept (see rules)
Blank member_casual	0	None needed
Final clean dataset	5,949,280	166,702 rows removed (2.7%)

Cleaning rules

- **Analysis period:** rides that started between Sep 1, 2025 and Aug 31, 2026. Divvy files are organized by ride end date, so rides crossing midnight at month-end fall into the next month's file.
- **Duration:** rides between 1 minute and 12 hours were kept. Rides under 1 minute were treated as false starts or re-docks. Rides of 12 hours or more were treated as system errors or unreturned bikes: 86% of them exceeded 24 hours, consistent with automatic closure by the system.
- **Duplicates:** 35 ride_ids appeared twice. Rides starting on the last day of a month and ending on the first day of the next appear in both monthly files; one record was kept.
- **Blank stations:** 100% of rides with a blank start station are electric bikes, which can be parked outside docking stations. These rides are real and were kept; they are excluded only from station-level analysis.
- **New fields:** ride length (seconds and minutes), day of week, start hour and month.

Validation

- September row count matched between Excel and BigQuery (714,759 rides in the file).
- Rides per month reconciled with the source files: for September, 714,759 in the file − 268 rides that started in August + 71 September rides found in the October file = 714,562 rides that started in September.
- Final row count reconciled with the expected removals; the 63-row difference is explained by rows meeting more than one exclusion rule.
- Full-year results were consistent with the September exploration in Excel (share of short rides, rider mix, weekly and hourly patterns).

4. Analyze

The clean table was aggregated in SQL by rider type and by month, day of week, start hour and station. Shares were calculated within each rider type (window functions), so the two groups can be compared despite their different sizes.

Metric (Sep 2025 – Aug 2026)	Casual	Member
Rides	2,079,522 (35%)	3,869,758 (65%)
Average ride length	17.8 min	12.0 min
Peak month / lowest month	Jul 345K / Jan 24K (14x)	Aug 511K / Dec 109K (5x)
Peak day (share of group's rides)	Saturday (21.1%)	Tue & Wed (16.0%)
Share of rides on weekdays	62.8%	76.7%
Weekday peak hours (share)	5 PM (10.8%), 8 AM (4.4%)	5 PM (11.9%), 8 AM (8.5%)

Summary of findings

1. Members ride more often; casual riders ride longer. Members took 65% of rides. Casual rides lasted ~48% longer on average (17.8 vs. 12.0 minutes), and their duration rises on weekends (20.3 min on Sunday vs. 14.8 min on Wednesday), while member duration stays stable (11.5–13.2 min).

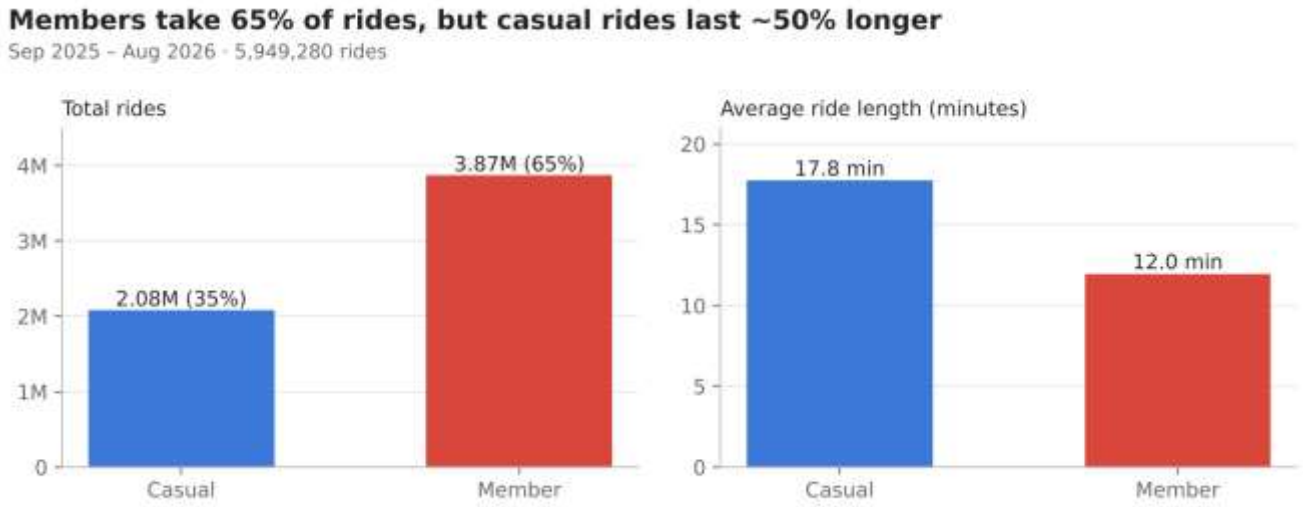
2. Casual ridership is far more seasonal. Casual rides fall 14x from July (345K) to January (24K); member rides fall 5x. The casual share of rides ranges from ~18% in January to ~41% in July. Casual ridership nearly triples from March (85K) to May (236K).

3. Members ride mid-week; casual riders peak on weekends, but most casual rides are on weekdays. Saturday is the casual peak (21.1% of casual rides), while members peak on Tuesday and Wednesday (16.0%). Still, 62.8% of casual rides (about 1.3 million) happen Monday to Friday. Per day, weekends are busier for casual riders; in total volume, weekdays dominate.

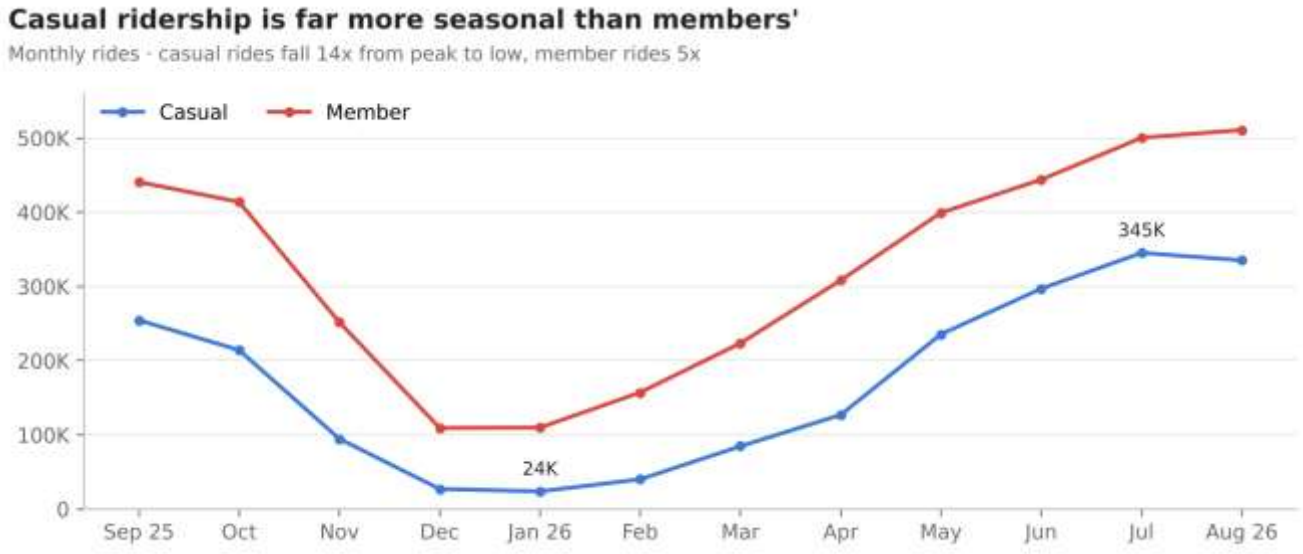
4. A segment of casual riders already commutes. On weekdays, both groups peak at 5 PM. Members also show a strong 8 AM peak (8.5%); casual riders show a smaller one (4.4%). The morning peak is the cleanest commuting signal; the 5 PM peak mixes commuting and after-work leisure. Between ~144K (7–9 AM only, conservative) and ~526K (7–9 AM plus 4–6 PM, upper bound) casual weekday rides fit a commuting pattern, 11% to 40% of casual weekday rides.

5. Commuting casual rides start at identifiable locations. In weekday peak hours overall, 7 of the top 10 casual start stations are tourist or lakefront locations. In the morning (7–9 AM), only 3 are; the rest are next to the Ogilvie and Union Station Metra terminals (3 stations) or in residential and office areas such as River North, Gold Coast and South Loop (4 stations).

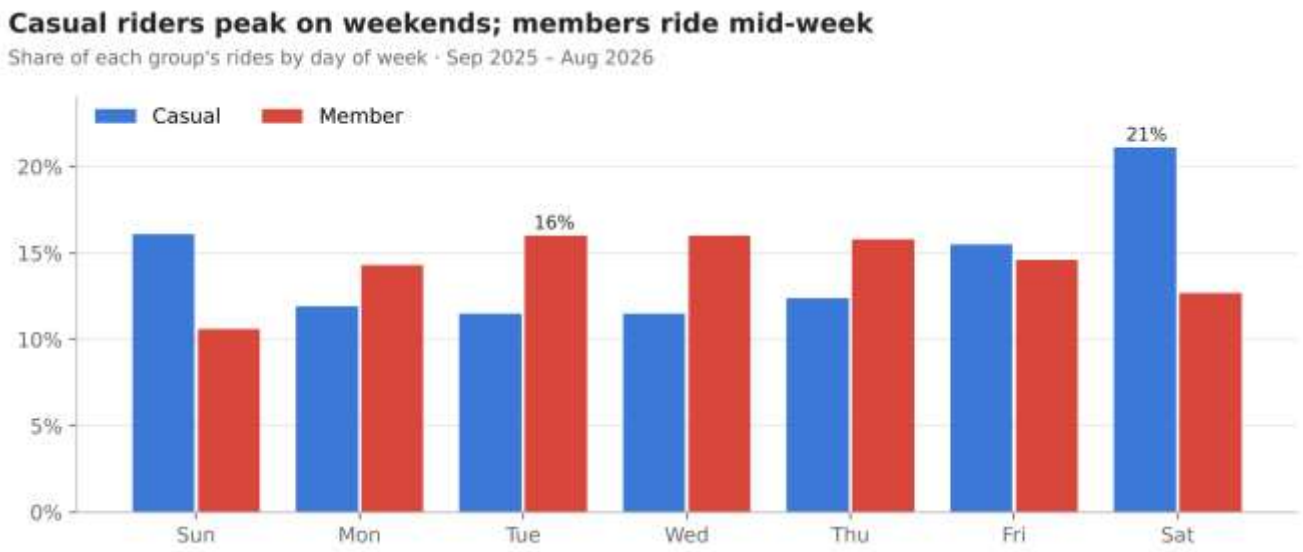
5. Share



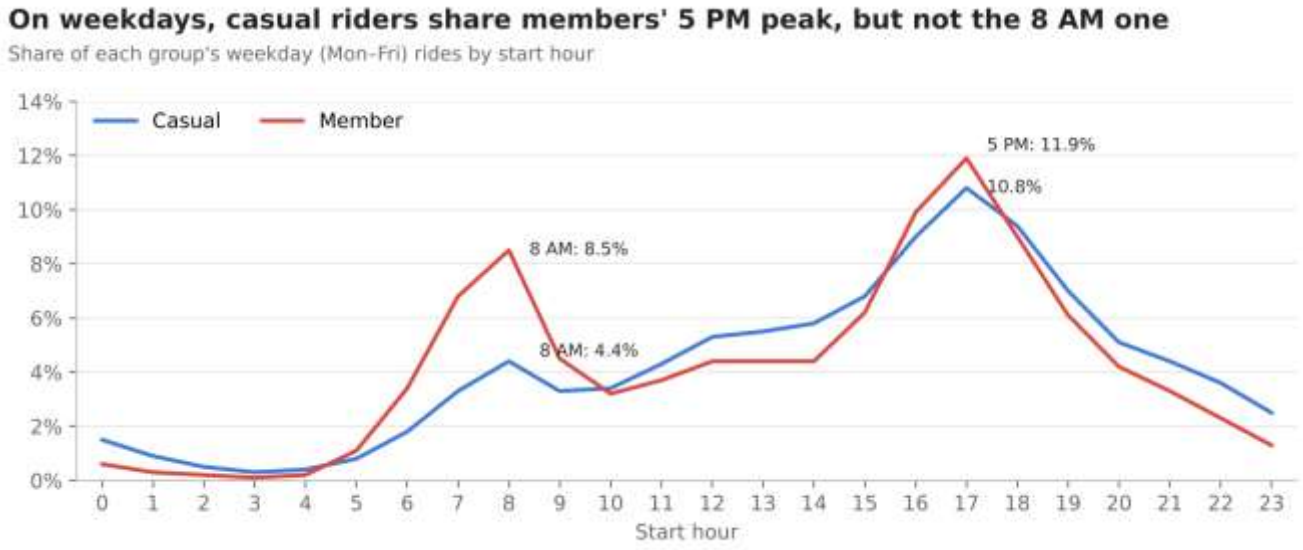
Members take most rides, but casual riders ride longer, consistent with more leisure use.



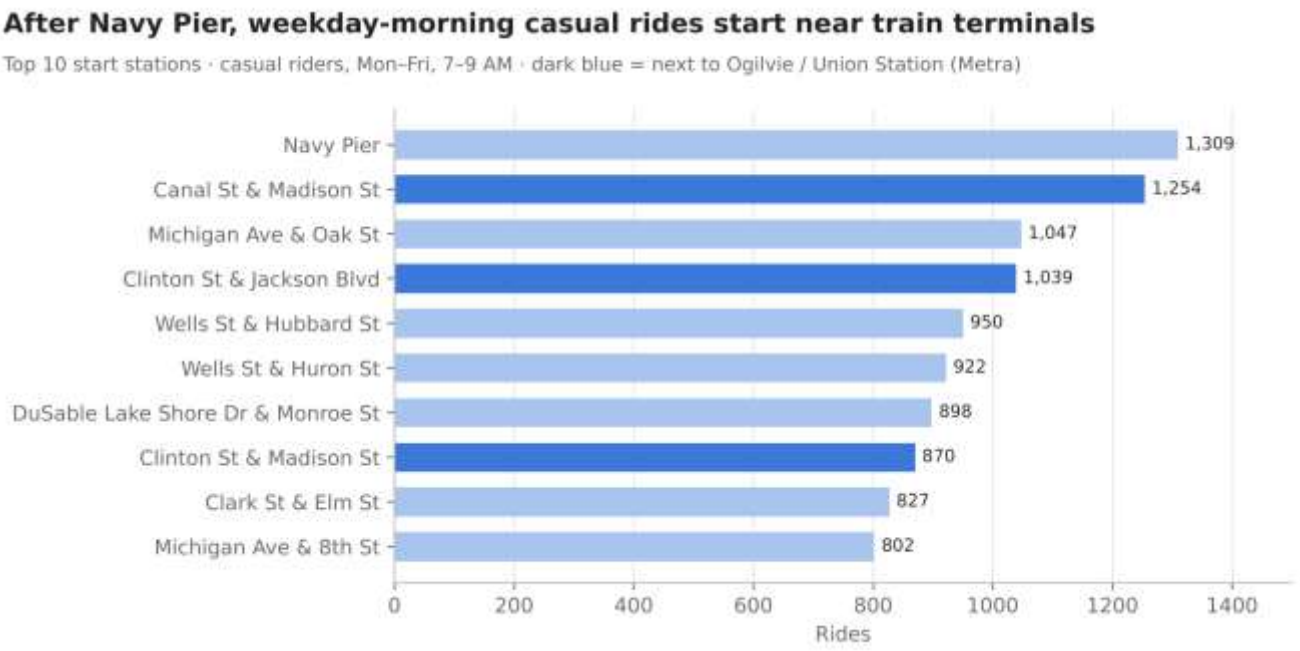
Both groups drop in winter, but casual ridership almost disappears. A campaign must reach casual riders while they ride: spring to early fall.



Opposite weekly patterns: members ride most mid-week, casual riders on weekends. Still, 63% of casual rides happen on weekdays.



Weekdays only. Members show the two commuting peaks; casual riders follow the 5 PM peak and show a smaller 8 AM peak, evidence that part of the casual base already commutes.



Weekday mornings isolate the cleanest commuting signal. Highlighted stations are next to the Ogilvie and Union Station Metra terminals; others in the list are mostly residential and office areas.

6. Act: top three recommendations

1. Target weekday commuters at Metra terminals and business districts

Who: casual riders starting weekday rides at 7–9 AM.

Where: stations next to Ogilvie and Union Station, plus River North, Gold Coast and South Loop.

How: digital, in-app messages triggered by rides that match this pattern, highlighting the value of membership for daily commuting.

Evidence: 144K–526K casual weekday rides fit a commuting pattern; 3 of the top 10 weekday-morning casual stations are next to Metra terminals.

Success metric: conversion rate of casual riders with repeated weekday peak-hour rides.

2. Run the campaign from spring to early fall, ahead of the winter drop

When: launch in March–April, before casual ridership nearly triples (85K in March to 236K in May); push hardest through the May–August peak; close conversion by October, before the decline. Casual riders who keep riding in winter are likely committed users and a high-priority segment.

Evidence: casual rides fall 14x from peak to low, versus 5x for members.

Success metric: new annual memberships per month, March–October.

3. Position membership as one pass for commuting and leisure

Who: weekend leisure casual riders.

Where and when: Navy Pier, lakefront and Millennium Park, on summer weekends.

Message: members also ride for leisure: about 23% of member rides happen on weekends, with longer trips.

Evidence: casual rides peak on Saturday (21.1%) and last ~48% longer than member rides.

Note: membership savings must be validated by Finance, as the dataset contains no pricing information.

Success metric: conversions among riders whose casual trips are mostly on weekends.

Next steps to strengthen the analysis

- Rider-level (anonymized) IDs would allow counting unique casual riders and their ride frequency, the best predictor of conversion.
- Pricing and pass-type data would allow calculating the break-even point where membership becomes cheaper than paying per ride.
- End-station analysis for afternoon rides would test whether casual riders return to Metra terminals in the evening.

Appendix A: Exploratory analysis (September 2025, Excel)

Before the full-year analysis, one month (695,646 valid rides) was explored in Excel to test quality checks and form hypotheses. Main results, all later confirmed with the full year:

- Members took 63% of rides; casual rides were ~50% longer (18.9 vs. 12.5 minutes).
- Members peaked mid-week (18.5% on Tuesday); casual riders on Saturday (18.1%) and Sunday (16.0%).
- Members showed 8 AM and 5 PM peaks; on weekdays casual riders also peaked at 5 PM, and the peak disappeared on weekends.

Limitations of this step: a single summer month; September 2025 had 5 Mondays and Tuesdays and included Labor Day (Monday, Sep 1), which inflated Monday ridership. These effects disappeared in the full-year data.

Appendix B: SQL queries

All queries were saved in BigQuery and in the project folder, numbered in execution order.

Query	Purpose
01_create_all_trips_raw	Combine the 12 monthly tables with UNION ALL
02_validate_rows_by_month	Rides per month, to check that no month is missing or duplicated
03_data_quality_checks	Duplicates and blank values
04_investigate_duplicates	Find duplicate ride_ids and their cause
05_check_rideable_type	Blank stations by bike type
06_create_all_trips_clean	Apply cleaning rules and create new fields
07_validate_clean_rows	Row count of the clean table
08_summary_by_member_type	Rides and average length by rider type
09_rides_by_month_member_type	Seasonality
10_rides_by_weekday_member_type	Day of week (counts, share, duration)
11_rides_by_hour_weekdays	Weekday hourly pattern
12_top_stations_casual_commute	Top casual start stations, weekday peak hours
13_top_stations_casual_morning	Top casual start stations, weekday 7–9 AM

Cleaning query (06)

```
CREATE OR REPLACE TABLE `cyclistic.all_trips_clean` AS
SELECT DISTINCT
  ride_id, rideable_type, member_casual, started_at, ended_at,
  start_station_name, end_station_name,
  start_lat, start_lng, end_lat, end_lng,
  TIMESTAMP_DIFF(ended_at, started_at, SECOND) AS ride_length_sec,
  ROUND(TIMESTAMP_DIFF(ended_at, started_at, SECOND) / 60, 1) AS ride_minutes,
  EXTRACT(DAYOFWEEK FROM started_at) AS day_of_week,
  EXTRACT(HOUR FROM started_at) AS start_hour,
  FORMAT_TIMESTAMP('%Y-%m', started_at) AS month
FROM `cyclistic.all_trips_raw`
WHERE started_at >= '2025-09-01'
  AND started_at <  '2026-09-01'
  AND TIMESTAMP_DIFF(ended_at, started_at, SECOND) >= 60
  AND TIMESTAMP_DIFF(ended_at, started_at, SECOND) <  43200;
```

Weekday share by rider type (10), using a window function

```
SELECT
  member_casual,
  day_of_week,
  COUNT(*) AS total_trips,
  ROUND(AVG(ride_minutes), 1) AS avg_ride_minutes,
  ROUND(100 * COUNT(*) / SUM(COUNT(*)) OVER (PARTITION BY member_casual), 1) AS pct_of_group
FROM `cyclistic.all_trips_clean`
GROUP BY member_casual, day_of_week
ORDER BY member_casual, day_of_week;
```